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SMA EXPERIMENT NO : 09

CODE:

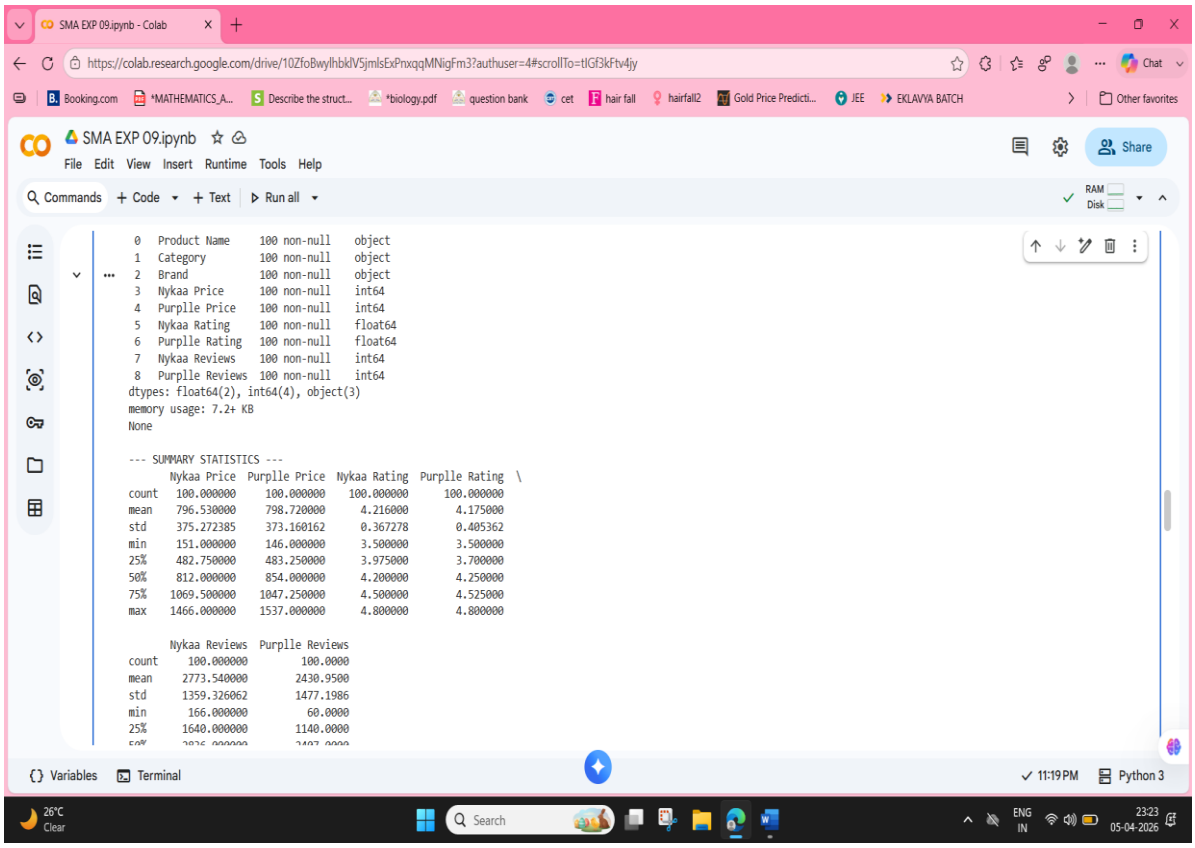
```
# ================================
# IMPORT LIBRARIES
# ================================
import pandas as pd
import matplotlib.pyplot as plt
# ================================
# LOAD DATASET
# ================================
df = pd.read_csv("/content/nykaa_purplle_dataset_100.csv")
print("Dataset Loaded Successfully\n")
print(df.head())
print("\nShape:", df.shape)
# ================================
# BASIC INFO
# ================================
print("\n--- DATASET INFO ---")
print(df.info())
print("\n--- SUMMARY STATISTICS ---")
print(df.describe())
# ================================
# PRICE ANALYSIS
# ================================
df["Price Difference"] = df["Nykaa Price"] - df["Purplle Price"]
avg_nykaa_price = df["Nykaa Price"].mean()
avg_purplle_price = df["Purplle Price"].mean()
print("\n--- PRICE ANALYSIS ---")
print("Average Nykaa Price:", avg_nykaa_price)
print("Average Purplle Price:", avg_purplle_price)
```

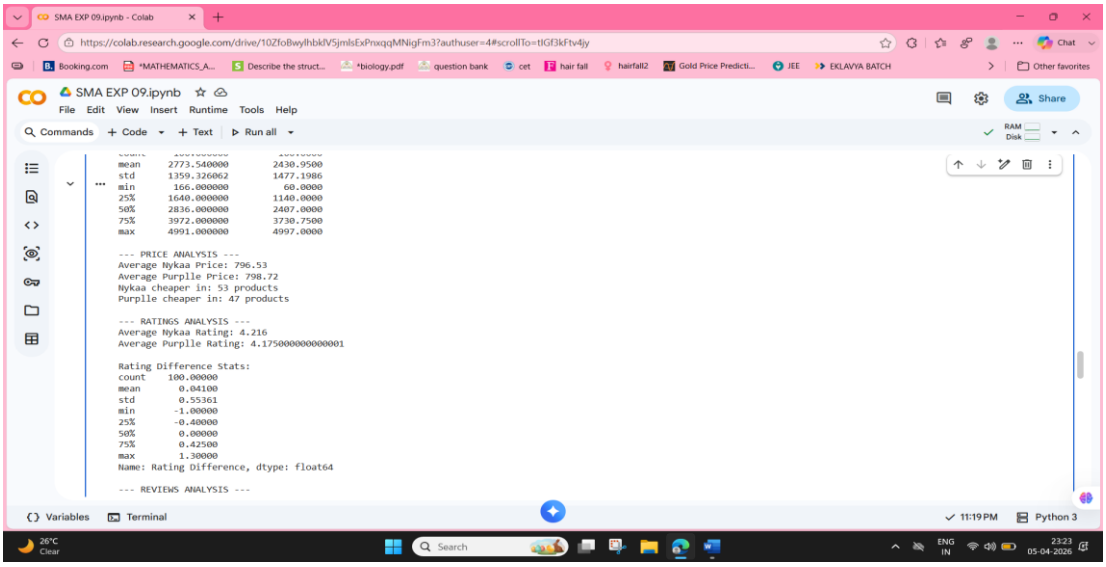
```
cheaper_nykaa = (df["Nykaa Price"] < df["Purplle Price"]).sum()
cheaper_purplle = (df["Purplle Price"] < df["Nykaa Price"]).sum()
print("Nykaa cheaper in:", cheaper_nykaa, "products")
print("Purplle cheaper in:", cheaper_purplle, "products")
# ================================
# RATINGS ANALYSIS
# ================================
avg_nykaa_rating = df["Nykaa Rating"].mean()
avg_purplle_rating = df["Purplle Rating"].mean()
df["Rating Difference"] = df["Nykaa Rating"] - df["Purplle Rating"]
print("\n--- RATINGS ANALYSIS ---")
print("Average Nykaa Rating:", avg_nykaa_rating)
print("Average Purplle Rating:", avg_purplle_rating)
print("\nRating Difference Stats:")
print(df["Rating Difference"].describe())
# ================================
# REVIEWS ANALYSIS
# ================================
avg_nykaa_reviews = df["Nykaa Reviews"].mean()
avg_purplle_reviews = df["Purplle Reviews"].mean()
total_nykaa_reviews = df["Nykaa Reviews"].sum()
total_purplle_reviews = df["Purplle Reviews"].sum()
print("\n--- REVIEWS ANALYSIS ---")
print("Average Nykaa Reviews:", avg_nykaa_reviews)
print("Average Purplle Reviews:", avg_purplle_reviews)
print("Total Nykaa Reviews:", total_nykaa_reviews)
print("Total Purplle Reviews:", total_purplle_reviews)
# ================================
# PLATFORM WINNER LOGIC
# ================================
def decide_winner(row):
    if row["Nykaa Price"] < row["Purplle Price"]:
```

```
        return "Nykaa"
    else:
        return "Purplle"
df["Cheaper Platform"] = df.apply(decide_winner, axis=1)
print("\n--- PLATFORM WINNER ---")
print(df["Cheaper Platform"].value_counts())
# ================================
# VISUALIZATIONS
# ================================
# Price Distribution
plt.figure()
plt.hist(df["Nykaa Price"], alpha=0.5, label="Nykaa")
plt.hist(df["Purplle Price"], alpha=0.5, label="Purplle")
plt.title("Price Distribution")
plt.xlabel("Price")
plt.ylabel("Count")
plt.legend()
plt.show()
# Ratings Distribution
plt.figure()
plt.hist(df["Nykaa Rating"], alpha=0.5, label="Nykaa")
plt.hist(df["Purplle Rating"], alpha=0.5, label="Purplle")
plt.title("Ratings Distribution")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.legend()
plt.show()
# Reviews Distribution
plt.figure()
plt.hist(df["Nykaa Reviews"], alpha=0.5, label="Nykaa")
plt.hist(df["Purplle Reviews"], alpha=0.5, label="Purplle")
plt.title("Reviews Distribution")
```

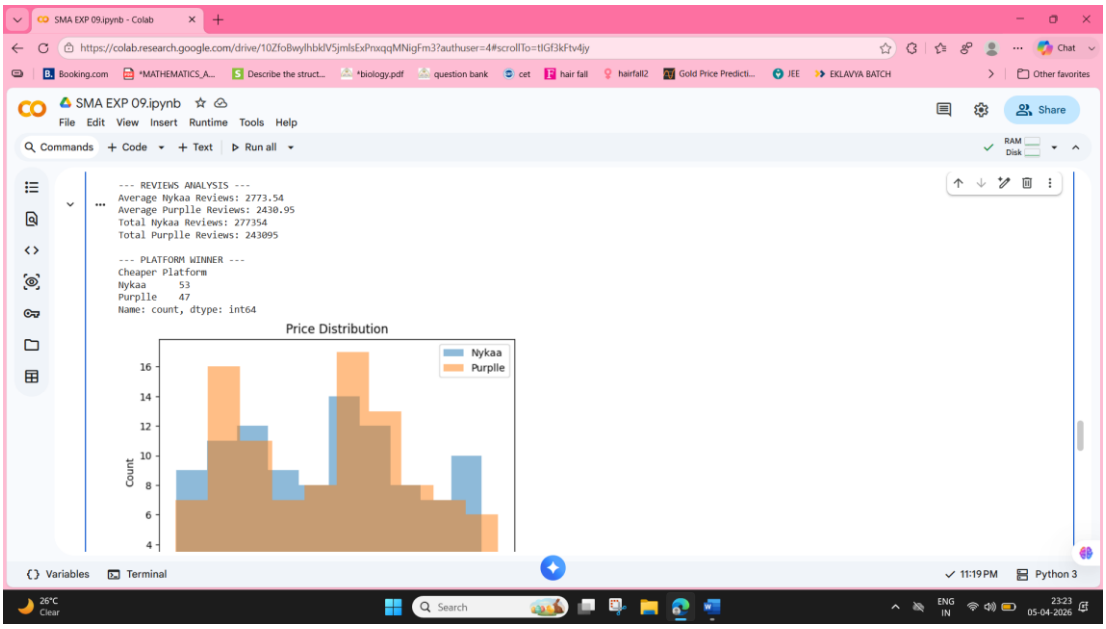
```
plt.xlabel("Reviews")
plt.ylabel("Count")
plt.legend()
plt.show()
# ================================
# FINAL INSIGHTS
# ================================
print("\n================ FINAL INSIGHTS ================")
print("Nykaa cheaper in:",
    (df["Nykaa Price"] < df["Purplle Price"]).sum(), "cases")
print("Purplle has higher rating in:",
    (df["Purplle Rating"] > df["Nykaa Rating"]).sum(), "cases")
print("Nykaa has more total reviews:",
    total_nykaa_reviews > total_purplle_reviews)
print("\nAnalysis Completed Successfully!")
```

OUTPUT:

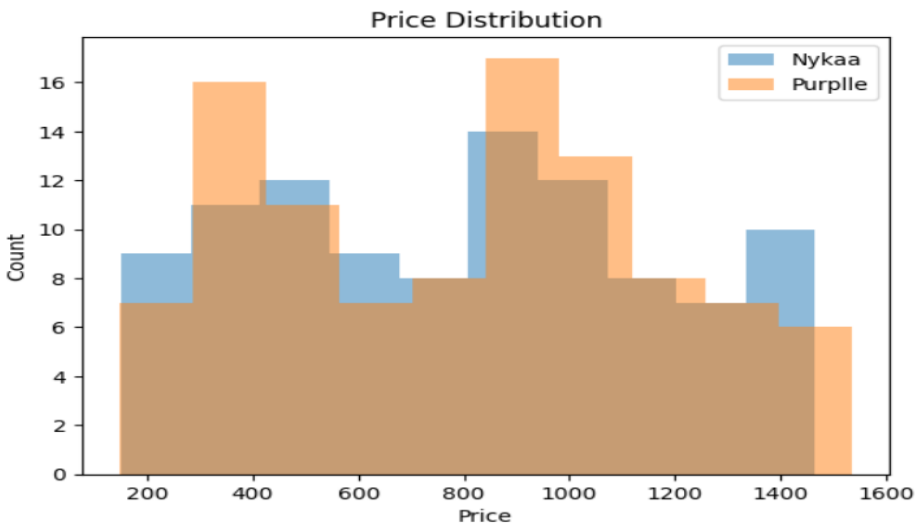


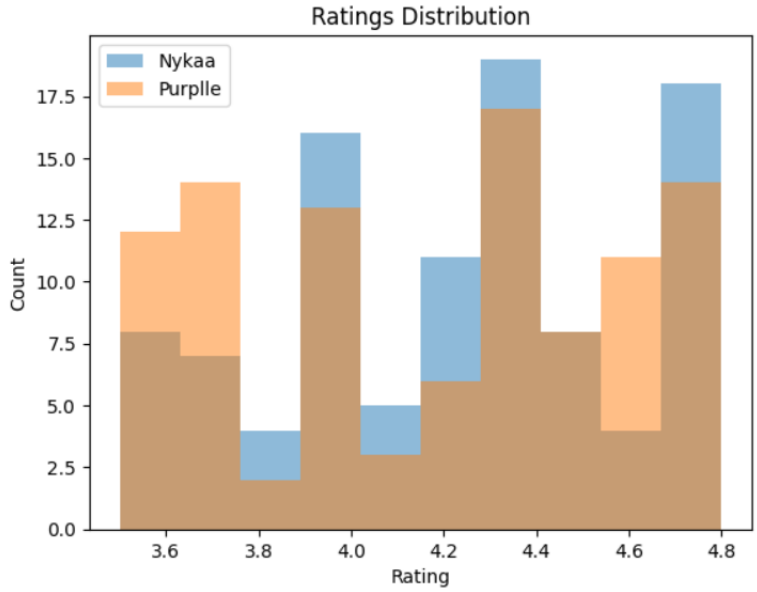


```
mean      2773.540000        2430.9500
std       1359.326062        1477.1986
min        166.000000          60.0000
25%       1640.000000        1140.0000
50%       2836.000000        2407.0000
75%       3972.000000        3730.7500
max       4991.000000        4997.0000

--- PRICE ANALYSIS ---
Average Nykaa Price: 796.53
Average Purplle Price: 798.72
Nykaa cheaper in: 53 products
Purplle cheaper in: 47 products

--- RATINGS ANALYSIS ---
Average Nykaa Rating: 4.216
Average Purplle Rating: 4.175000000000001

Rating Difference Stats:
count    100.00000
mean       0.04100
std        0.55361
min       -1.00000
25%       -0.40000
50%        0.00000
75%        0.42500
max        1.30000
Name: Rating Difference, dtype: float64

--- REVIEWS ANALYSIS ---
```



```
--- REVIEWS ANALYSIS ---
Average Nykaa Reviews: 2773.54
Average Purplle Reviews: 2430.95
Total Nykaa Reviews: 277354
Total Purplle Reviews: 243095

--- PLATFORM WINNER ---
Cheaper Platform
Nykaa      53
Purplle    47
Name: count, dtype: int64
```


Ratings Distribution
Nykaa
Purplle
Count
Rating
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
3.6
3.8
4.0
4.2
4.4
4.6
4.8

Reviews Distribution
Nykaa
Purplle
Count
Reviews
0
2
4
6
8
10
12
14
16
0
1000
2000
3000
4000
5000

SMA EXP 09.ipynb - Colab
SMA EXP 09.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
Share
Count
Reviews
---------- FINAL INSIGHTS ----------
Nykaa cheaper in: 53 cases
Purplle has higher rating in: 43 cases
Nykaa has more total reviews: True
Analysis Completed Successfully!
Variables Terminal
Python 3